

Safe Adaptive Context and TACER: Task-Aware Evidence Routing for Token-Efficient Retrieval-Augmented Generation

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Abstract

Retrieval-Augmented Generation (RAG) pipelines often pass a fixed number of retrieved documents to the generator, even though evidence needs vary by query. We study context selection as a post-retrieval control problem: given the same candidate list, how much evidence should enter the prompt, and in what form? We present Safe Adaptive Context, a lightweight controller that starts with limited evidence and expands when the generated answer appears weak, and TACER, a task-aware extension that routes between compact evidence and safer broader context using retrieval-side and evidence-coverage signals. Across five datasets, two generators, and stronger retriever settings, the controllers substantially reduce provider-reported token use while preserving most Fixed Top-10 answer quality. In a separate 500-query external-baseline evaluation (100 queries per dataset), Safe Adaptive Context retains 99.8% of Fixed Top-10 F1 with 55.0% token reduction, while TACER retains 99.4% with 60.4% token reduction. These are complementary operating points rather than evidence that TACER universally improves quality.

1 Introduction

RAG improves language-model answers by retrieving external evidence and placing it in the prompt (Lewis et al., 2020). The dominant pattern is simple: retrieve a ranked list and pass a fixed number of documents, usually the top- k , to the generator. This treats context depth as a global constant rather than a query-level decision. A biomedical factoid question may need one sentence, a scientific claim may need a small evidence span, a multi-hop question may require several documents, and an ambiguous long-form question may require broader synthesis.

Fixed retrieval depth is inefficient in two ways. It spends unnecessary prompt tokens on easy or

concentrated-evidence queries, and it can reduce answer quality by placing useful evidence among distractors. Prior work shows that large language models can be distracted by irrelevant context (Shi et al., 2023), and that long-context factuality depends on how useful information is positioned and attended to (Wu et al., 2025). Token-efficient RAG is therefore not only a cost problem; it is also a context-control problem.

We study context selection as a post-retrieval decision. A stronger retriever can improve candidate order, but it still does not decide whether the generator should see one compact span, three full documents, sentence neighborhoods from several documents, or a broader fallback context. Our central finding is that retrieval-score concentration is not a reliable proxy for answer-level evidence sufficiency: adapting k alone can under-select evidence, especially for distributed and multi-hop tasks.

This distinction matters for grounded generation. A retrieved list can contain the right evidence while still presenting it in a form that is too sparse, too noisy, or too compressed for the generator. We therefore treat context construction as a policy decision over already retrieved evidence, rather than as a side effect of retriever quality.

Safe Adaptive Context addresses this by combining a lightweight budget predictor, evidence compression, and answer-aware fallback. It first attempts a smaller context and expands only when the generated answer looks weak or unsupported. TACER extends this idea by making the policy explicitly task-aware: it routes between compact and safer context construction depending on whether evidence appears concentrated or distributed, and whether the query likely requires short factual extraction, multi-hop reasoning, or longer synthesis.

TACER is not a new retriever or standalone prompt compressor; it is a lightweight post-retrieval controller that jointly chooses context

depth, evidence form, and fallback risk. We evaluate on five datasets, two generators, and three retrieval conditions. Our comparisons include fixed top- k , heuristic score-gap rules, Adaptive- k -style score baselines, LLMLingua-2, and a constrained FLARE-style baseline. Code and materials are publicly available.

2 Related Work

Retrieval and Context Allocation. RAG conditions generation on retrieved evidence (Lewis et al., 2020); open-domain QA systems have long used retrieval for grounding, including dense retrieval methods such as DPR (Karpukhin et al., 2020). Retrieval benchmarks such as BEIR evaluate ranking quality across domains (Thakur et al., 2021). Most RAG pipelines, however, still require a context-construction step after retrieval. Better retrieval increases the chance that useful evidence appears in the candidate set, but a high-quality top-10 list can still contain redundant passages, and a high-scoring first passage can still be insufficient for multi-hop reasoning.

Adaptive Retrieval and Compression. Several approaches study when to retrieve or compress more. FLARE retrieves when generation appears uncertain (Jiang et al., 2023b), Self-RAG trains models to critique retrieval and generation (Asai et al., 2024), and Adaptive- k selects the number of passages from the query-passage similarity distribution without fine-tuning or extra LLM calls (Taguchi et al., 2025). Prompt-compression methods such as LLMLingua, LLMLingua-2, Selective Context, and RECOMP reduce context length by selecting, deleting, or rewriting text (Jiang et al., 2023a; Pan et al., 2024; Li et al., 2023; Xu et al., 2024). EXIT and AdaComp are especially relevant because they adapt extractive compression to query and retrieval signals (Hwang et al., 2025; Zhang et al., 2024).

TACER is complementary to these methods. It does not train a compression model, call an LLM before generation, or replace retrieval. Instead, it treats compression as one action available to a controller: use compact evidence when it appears safe, preserve broader context when evidence is distributed, and fall back when the first answer appears unsupported.

Reranking, Distractors, and Positioning. Cross-encoder reranking and dense retrieval

improve candidate ordering, but they do not decide how much context the generator should receive. Dense retrievers such as DPR learn semantic query-passage matching (Karpukhin et al., 2020), while cross-encoder rerankers often improve top-rank precision (Nogueira and Cho, 2019); we include both TF-IDF+cross-encoder and dense+cross-encoder settings to test whether context routing remains useful after ranking improves. This matters because irrelevant or poorly placed context can harm generation when useful evidence is mixed with distractors (Shi et al., 2023), and long-context factuality depends on how retrieval-related attention behaves (Wu et al., 2025). The closest prior work covers parts of our system: Adaptive- k selects a cutoff, LLMLingua-style methods compress prompts, FLARE retrieves more when generation appears uncertain, and EXIT/AdaComp adapt compression. Our contribution is the combination in a low-overhead post-retrieval controller that chooses compact evidence, broader context, and answer-aware fallback as policy actions.

3 Method

3.1 Problem Setup

For a query q , a retriever returns ranked candidates $R_q = [(d_1, s_1), \dots, (d_{10}, s_{10})]$, where d_i is the document at rank i and s_i is the retrieval score. A context policy transforms this list into prompt context C_q , and the generator produces answer a_q from q and C_q . All methods in a retrieval condition share the same candidate list; only the context policy changes.

This controlled setup isolates the question we want to study. If two methods receive the same ranked evidence pool and use the same generator, differences in quality and token cost reflect context allocation rather than retrieval coverage, model choice, or decoding randomness.

We measure cost as provider-reported total tokens used to obtain the final answer. For fallback methods, this includes both the first attempt and the fallback generation, because a failed first pass still costs tokens in real use.

3.2 Context Policies

Fixed Top- k baselines pass Top-3, Top-5, or Top-10 full documents. Fixed Top-10 is the expensive reference point for token-reduction calculations, but it is not a guaranteed quality upper bound be-

cause longer prompts can include distractors. The Heuristic Rules baseline uses retrieval-score gaps and query length to select a budget, but does not compress evidence or inspect answer risk.

We distinguish two Adaptive- k -style baselines. *Adaptive- k (score-drop)* is our lightweight normalized largest-adjacent-drop baseline over the shared top-10 candidate list; it is used in the main and retriever-robustness experiments. *Adaptive- k (paper)* is a paper-faithful reproduction of the published largest-gap rule, used in the external-baseline evaluation and blinded annotation. Both are strong lightweight baselines because they require no training, no extra LLM call, and no answer-time critique. Their shared limitation is that score shape does not always encode evidence need: a multi-hop question can require lower-ranked evidence even when the first score gap is large.

We keep these baselines separate because they answer different reviewer questions. The score-drop variant is a transparent in-harness diagnostic over the same top-10 pool as TACER, while the paper-faithful variant checks whether the claim survives against a closer reproduction of prior adaptive- k behavior.

Safe Adaptive Context predicts an initial budget from retrieval-side features, constructs compact evidence spans when appropriate, generates once, and expands context if a surface-level risk checker detects an empty, weakly grounded, or unsupported answer. TACER extends this controller with task-aware routing. It estimates whether the evidence is concentrated or distributed and chooses between aggressive compact evidence, safer selected context, or fallback-enabled expansion. Table 1 summarizes the routing signals and actions.

For compact evidence, selected documents are split into sentences and scored using lexical overlap, rare query-term matches, phrase overlap, token density, and a small position bonus. TACER keeps high-scoring sentence neighborhoods rather than isolated sentences, preserving nearby definitions, negations, and conditions. The exact routing thresholds are reported in the appendix and released code.¹

The final routing policy uses only query text, retrieval scores, and lexical evidence coverage; it

does not use dataset identifiers or an additional LLM call before generation. Concise prompts with bridge evidence, multi-hop shape, or distributed complexity use the safer Safe Adaptive ladder. Long-answer prompts with more than 120 output tokens and non-focused complexity also use the safer ladder. Claim-style prompts with at most 120 output tokens use compact coverage-guided evidence unless evidence is distributed. Focused or concentrated evidence otherwise uses compact coverage-guided evidence. Evidence complexity is assigned from general query features: entity count, connective cues, content-word length, and whether query evidence appears across several documents.

Fallback is deliberately conservative. Evidence-style prompts fall back for empty answers, weak-answer phrases, low answer-context overlap, low context-query coverage, very short unsupported answers, or suspiciously verbose answers with poor grounding. Concise short-answer prompts fall back only for empty answers or unusually long answers with weak answer-context overlap, because correct answers can be names, dates, places, or short phrases with limited lexical overlap.

Computational cost. TACER adds little overhead beyond retrieval and generation. Its routing features are computed from the already available ranked list and simple lexical evidence scoring. Unlike methods that ask an LLM to classify the query before generation, TACER does not introduce a separate pre-generation model call. The only additional LLM cost occurs when answer-aware fallback is triggered, and fallback cost is included in all reported token totals.

4 Experimental Setup

We evaluate on SciFact, BioASQ, HotpotQA, MS MARCO, and ASQA, covering scientific claim verification, biomedical factoid QA, multi-hop QA, web passage QA, and ambiguous long-form QA. Table 2 summarizes the dataset roles. Each dataset uses a 50-example calibration split for lightweight routing choices and a fixed 200-example evaluation split. The main generator is meta-llama/Llama-3.3-70B-Instruct; a local Ollama Mistral 7B model is used as a smaller-model robustness check. The main retrieval setting is TF-IDF. Reranking uses cross-encoder/ms-marco-MiniLM-L-6-v2;

dense	retrieval	uses
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¹<https://github.com/Joel-Vazquez-Lopez/adaptive-rag-token-efficiency>

Table 1: TACER routing signals and actions. The policy is task-aware but not dataset-specific.

Signal family	Features	Routing role
Retrieval concentration	Top scores, score gaps, score mass, entropy, normalized entropy	Detect focused vs distributed ranking structure
Evidence coverage	Query length, lexical overlap, phrase coverage, useful spans across documents	Decide whether compact evidence covers the query
Answer-form risk	Factoid/claim-style cues, multi-hop cues, long-form or explanatory query shape	Choose compact evidence vs safer broader context
Fallback risk	Empty answer, weak answer phrase, low answer-context overlap, low query coverage	Expand context after an unsupported first pass

intfloat/multilingual-e5-large-instruct followed by the same reranker.

The main evaluation uses 200 held-out queries per dataset (1,000 total). A separate external-baseline evaluation uses 100 queries per dataset (500 total); all methods within an evaluation use the same query set. SciFact answer-level metrics use auxiliary natural-language references generated from the claim and gold evidence with openai/gpt-oss-120b at temperature 0. The prompt requires a concise answer based only on the supplied gold evidence. These references are used only for answer-level metrics; retrieval diagnostics use the original relevance labels. The other datasets use references prepared from their source data or simplified JSONL conversions. Full prompts and preparation details are provided in the appendix.

The retriever-robustness conditions are not intended as a retrieval benchmark. They ask whether context allocation remains useful after candidate ordering improves. Final reported runs require provider-reported token counts, and all generators run at temperature 0 to make method comparisons paired and reproducible.

Table 3 summarizes the compared methods. The external evaluation adds three stronger alternatives beyond the original Safe Adaptive evaluation. *Adaptive-k (paper)* uses fixed buffer $B = 5$ and top-90% gap search, adapted to our top-10 candidate lists. LLMLingua-2 uses its MeetingBank checkpoint to compress either Fixed Top-10 or *Adaptive-k (paper)* evidence using structured document blocks; the exact identifier is given in the appendix. FLARE-lite is a constrained FLARE-style baseline: it starts with minimal evidence and expands when the answer appears weak. We do not present FLARE-lite as official FLARE, because the hosted API setting does not expose the token-level uncertainty signal used by the original method.

Metrics. We evaluate answer quality with token F1, answer coverage, and semantic similarity. Coverage is reference-token recall, $|T(a) \cap T(r)|/|T(r)|$, for normalized

unique-token sets $T(\cdot)$. Semantic similarity is cosine similarity between answer and reference embeddings from sentence-transformers/all-MiniLM-L6-v2 (Reimers and Gurevych, 2019). Efficiency is measured with provider-reported total tokens, including both generations after fallback, and token reduction relative to Fixed Top-10 under the same dataset, model, and retriever. Conventional retrieval nDCG@10 and MRR@10 are fixed for a retriever and dataset; when these diagnostics vary by context policy, we label them Ctx nDCG and Ctx MRR because they measure relevance retained in the selected prompt context, not retriever quality.

For dataset d and method m , F1 retention is $100 \times F1_{d,m}/F1_{d,Top10}$, and token reduction is $100 \times (1 - Tokens_{d,m}/Tokens_{d,Top10})$. Cross-dataset summaries macro-average these dataset-level retention and reduction values, so displayed average retention need not equal the ratio of displayed average F1 columns.

5 Results

5.1 Main Results

Table 4 reports the main Llama-70B TF-IDF results. TACER is highly efficient on concentrated-evidence datasets and avoids the largest *Adaptive-k* (score-drop) failures on HotpotQA and ASQA. The clearest failure is HotpotQA: *Adaptive-k* (score-drop) reduces token use by 70.4%, but F1 drops from 0.645 to 0.490. TACER spends more context, saves 32.6%, and recovers most answer quality with F1 0.627. On SciFact, where evidence is often concentrated, TACER reaches the highest F1 while reducing tokens by 80.9%. On ASQA, TACER gives up some token savings relative to *Adaptive-k* (score-drop) but improves F1 by 0.036.

5.2 Component Ablation

The main tables show final operating points but do not isolate why the controller helps. Table 5 reports a small SciFact ablation for the Safe Adaptive pipeline. Compact evidence is the main efficiency driver: Adaptive Compact Only preserves

Table 2: Datasets used in the evaluation. The split is fixed across methods so token and answer-quality comparisons are paired by query.

Dataset	Domain	Task type	Evidence pattern	Calib.	Eval.
SciFact	Scientific/biomedical	Claim verification	Concentrated evidence	50	200
BioASQ	Biomedical	Factoid QA	Focused technical evidence	50	200
HotpotQA	General-domain	Multi-hop QA	Distributed evidence	50	200
MS MARCO	Web passages	Passage QA	Often early-answer evidence	50	200
ASQA	General-domain	Ambiguous long-form QA	Explanatory/synthesis evidence	50	200

Table 3: Compared context-selection methods.

Method	Budget signal	Compression	Fallback	Task/evidence awareness
Fixed Top-3/5/10	Fixed constant	No	No	No
Heuristic Rules	Score gaps + query length	No	No	Weak
Adaptive- k (score-drop)	Score distribution	No	No	No
LLMLingua-2	Fixed or paper Adaptive- k context	Model-based	No	Compression only
FLARE-lite	Answer-risk expansion	No	Yes	Weak
Safe Adaptive Context	Predicted budget + answer risk	Yes	Yes	Moderate
TACER	Score shape + evidence coverage + task routing	Yes	Yes	Stronger

Fixed Top-10 F1 while reducing tokens by 77.9%. Fallback changes the risk profile: the full Safe Adaptive pipeline preserves F1 with a 76.5% token reduction and a 4.0% fallback rate. This motivates TACER’s routing layer, which decides when compact evidence is appropriate rather than applying compression uniformly.

5.3 External Baselines and Frontier

Table 6 reports the separate 500-query external-baseline evaluation, comprising 100 queries from each dataset. At the single tested operating point, Adaptive- k (paper) preserves quality with 24.9% average token reduction. The two LLMLingua-2 configurations compress more aggressively and retain roughly 80–82% of Fixed Top-10 F1. Because LLMLingua-2 compression was not swept, these results compare the reported configurations rather than establish a complete Pareto frontier. Safe Adaptive Context and TACER retain nearly all Fixed Top-10 F1 while reducing tokens by 55.0% and 60.4%, respectively.

5.4 Complementary Quality Metrics

Token F1 alone can be misleading because correct paraphrases may share few words with the reference, while short generic answers can receive some overlap. Table 7 therefore reports coverage and semantic similarity for the main TF-IDF setting. The complementary metrics reinforce the main interpretation. On HotpotQA, Adaptive- k (score-drop) drops not only in F1 but also in coverage and semantic similarity, while TACER recov-

ers most of both. On ASQA, TACER improves over Adaptive- k (score-drop) in all three quality metrics. BioASQ and MS MARCO remain more mixed, showing that aggressive routing can trade away some coverage even when F1 remains competitive.

5.5 Paired Confidence Check

Because all methods answer the same evaluation queries, Table 8 compares TACER directly against Adaptive- k (score-drop) with paired bootstrap intervals in the main TF-IDF setting. TACER’s F1 gains on HotpotQA and ASQA are clearly positive; SciFact, BioASQ, and MS MARCO are closer in F1 but generally favor TACER in token use. The confidence check makes the trade-off explicit: TACER spends more tokens where quality improves, and saves tokens where F1 is statistically closer.

5.6 Robustness and Human Evaluation

The pattern persists under stronger retrieval. On HotpotQA, Adaptive- k (score-drop) retains only 76.0% of Fixed Top-10 F1 under TF-IDF, 79.7% with TF-IDF+CE, and 79.1% with dense+CE, while TACER retains 97.3%, 96.3%, and 97.2%, respectively. This suggests the failure is not simply weak retrieval: multi-hop questions often require more evidence than the retrieval-score curve suggests.

Table 10 summarizes average behavior across retriever settings. Under stronger retrievers, Safe Adaptive matches or exceeds TACER’s aggregate

Table 4: Main Llama-70B TF-IDF results. Reductions are relative to Fixed Top-10.

Dataset	Fixed F1	Heuristic F1 / Red.	Score-drop F1 / Red.	Safe F1 / Red.	TACER F1 / Red.
SciFact	0.243	0.245 / 37.1%	0.251 / 74.9%	0.247 / 69.9%	0.253 / 80.9%
BioASQ	0.328	0.325 / 39.2%	0.319 / 73.2%	0.323 / 74.6%	0.314 / 80.8%
HotpotQA	0.645	0.575 / 37.3%	0.490 / 70.4%	0.627 / 32.3%	0.627 / 32.6%
MS MARCO	0.196	0.201 / 41.8%	0.199 / 48.6%	0.195 / 38.5%	0.195 / 54.1%
ASQA	0.422	0.411 / 44.0%	0.368 / 70.9%	0.402 / 58.3%	0.404 / 59.4%

Table 5: Component ablation on SciFact with Llama-70B. Reductions are relative to Fixed Top-10.

Variant	F1	Semantic	Tokens	Red.
Fixed Top-10	0.247	0.489	3325	0.0%
Adaptive Compact Only	0.248	0.481	736	77.9%
Adaptive Full Only	0.240	0.485	1097	67.0%
Top-5 Compact + Fallback	0.258	0.489	1209	63.6%
Full Safe Adaptive	0.247	0.477	780	76.5%

F1 retention and also saves more tokens. In the main TF-IDF run, paired TACER–Safe F1 intervals include zero on every dataset; TACER nevertheless uses fewer mean tokens on all five datasets. Thus TACER is an aggressive operating point in the main setting, not a uniformly superior controller.

The smaller Mistral generator shows the same qualitative pattern rather than a model-specific artifact. On HotpotQA, Adaptive- k (score-drop) drops to F1 0.282, while Safe Adaptive Context and TACER recover to 0.369 and 0.374 with roughly 31% token reduction. On SciFact and BioASQ, TACER remains highly token-efficient, reducing tokens by 79.4% and 79.8%, respectively. The appendix reports the full Llama–Mistral comparison.

We also conduct a small blinded annotation on HotpotQA and ASQA, comparing Fixed Top-10, Adaptive- k (paper), Safe Adaptive Context, and TACER using saved Llama-70B generations from the external-baseline evaluation. Systems are randomized as A/B/C/D within each query, and outputs are judged for factual correctness and grounding against the reference and selected evidence. Two annotators label all 120 outputs. Exact four-way agreement is 68.3% with Cohen’s $\kappa = 0.422$; binary acceptable/unacceptable agreement is 92.5% with $\kappa = 0.688$.

The annotation results are intentionally interpreted as a targeted grounding check, not a large-scale human evaluation. The sampled queries focus on HotpotQA and ASQA because they are where score-only context depth is most likely to

under-select evidence. In this setting, Adaptive- k (paper) has the highest rate of incorrect or not-enough-information judgments, while Safe Adaptive Context and TACER have higher acceptable rates under the conservative merge. The effect is concentrated on HotpotQA, where distributed evidence makes score-only selection risky. The result supports the main claim qualitatively: the proposed controllers do not merely optimize automatic F1; they reduce the risk of unsupported or incomplete answers in the cases where context routing should matter most. The appendix reports the split by dataset and the annotation protocol.

The agreement pattern is also informative. Exact labels differ partly because annotators often draw the boundary between correct and partially correct differently, but the acceptable/unacceptable agreement is high. This is the distinction most relevant to our claim: whether the selected context supports an answer well enough to be usable.

5.7 Human Annotation as Metric Validation

The targeted TACER comparison is not the only human signal. We also use the structured annotation study from the Safe Adaptive evaluation to validate how automatic metrics relate to human factual judgments. That study contains 120 outputs from SciFact, HotpotQA, and BioASQ, sampled across Llama-70B and Mistral. Two annotators agreed on 84 examples, with 70.0% raw agreement and Cohen’s $\kappa = 0.477$; among agreed labels, 65 were correct, 6 partially correct, and 13 incorrect. The appendix gives the dataset-level label counts.

The annotation confirms that token F1 is useful but incomplete. High F1 is strongly associated with human-correct answers, but low-F1 examples are mixed rather than uniformly wrong; acceptable paraphrases can have low lexical overlap. Semantic similarity shows the same pattern: high semantic similarity usually indicates acceptable answers, but medium scores are noisier and can over-credit

Table 6: External-baseline evaluation over 500 queries (100 per dataset) with Llama-70B. Token reduction and F1 retention are computed per dataset relative to Fixed Top-10, then macro-averaged.

Method	Avg F1	Avg Coverage	Avg Semantic	Avg Tokens	Avg Token Red.	Avg F1 Ret.
Fixed Top-10	0.366	0.608	0.651	2220	0.0%	100.0%
Adaptive- k (paper)	0.356	0.603	0.642	1643	24.9%	99.1%
Top-10 + LLMingua-2	0.280	0.494	0.584	622	70.9%	81.5%
Adaptive- k (paper) + LLMingua-2	0.275	0.491	0.582	492	76.6%	80.5%
FLARE-lite	0.335	0.515	0.609	708	66.4%	95.6%
Safe Adaptive Context	0.360	0.600	0.640	852	55.0%	99.8%
TACER	0.359	0.573	0.636	736	60.4%	99.4%

Table 7: Complementary quality metrics in the main Llama-70B TF-IDF setting. Reductions are relative to Fixed Top-10.

Dataset	Method	F1	Coverage	Semantic	Token Red.
SciFact	Top-10 / Score-drop / TACER	0.243 / 0.251 / 0.253	0.815 / 0.818 / 0.814	0.548 / 0.545 / 0.554	0.0 / 74.9 / 80.9%
BioASQ	Top-10 / Score-drop / TACER	0.328 / 0.319 / 0.314	0.507 / 0.458 / 0.464	0.753 / 0.738 / 0.740	0.0 / 73.2 / 80.8%
HotpotQA	Top-10 / Score-drop / TACER	0.645 / 0.490 / 0.627	0.651 / 0.493 / 0.630	0.747 / 0.605 / 0.724	0.0 / 70.4 / 32.6%
MS MARCO	Top-10 / Score-drop / TACER	0.196 / 0.199 / 0.195	0.614 / 0.549 / 0.523	0.473 / 0.466 / 0.456	0.0 / 48.6 / 54.1%
ASQA	Top-10 / Score-drop / TACER	0.422 / 0.368 / 0.404	0.419 / 0.337 / 0.382	0.780 / 0.730 / 0.764	0.0 / 70.9 / 59.4%

domain-related but incomplete responses. This is why we report F1, coverage, semantic similarity, and the targeted grounding annotation jointly rather than treating any single automatic metric as decisive.

5.8 Failure-Mode Analysis

The aggregate results suggest three recurring regimes. First, *score-only under-selection*: Adaptive- k (score-drop) selects a small prefix because the score curve has a clear drop, but the task still requires later evidence. HotpotQA is the clearest example. Second, *over-compression*: compact evidence can remove distractors, but also conditions, negations, or secondary facts; BioASQ and ASQA are more exposed to this risk. Third, *metric mismatch*: acceptable paraphrases can have low token overlap, while incomplete biomedical answers can receive high semantic similarity.

6 Discussion

The main result is that context selection is not only a budget problem. Adaptive- k -style score baselines are strong when the retrieval-score curve reflects evidence need, but they can fail when evidence is distributed. TACER targets this gap: it spends more tokens on multi-hop or explanatory queries and compresses more aggressively when evidence is localized. Fixed baselines show the same point from another angle: Top-3 and Top-5 can be competitive on MS MARCO and BioASQ, while Top-10 remains important on HotpotQA. Choosing a fixed k is already a hidden policy deci-

sion; TACER makes that policy explicit and query-dependent.

The external-baseline and annotation results keep the claim narrow. Safe Adaptive Context is preferable when quality preservation matters most; TACER can provide greater savings in the main TF-IDF setting; and score-only selection may suffice for concentrated, low-risk distributions.

What TACER adds. Beyond Adaptive- k , TACER adapts the context policy from expected evidence structure rather than only context size from the score curve. This lets it sacrifice token savings on HotpotQA to preserve quality, while compressing more aggressively on localized-evidence tasks. Beyond Safe Adaptive Context, TACER makes the routing decision explicit: it identifies when aggressive compression is appropriate and when broader context is needed. The two methods are complementary operating points rather than mutually exclusive alternatives.

7 Conclusion

We presented Safe Adaptive Context and TACER for token-efficient RAG. Across five datasets, two generator scales, stronger retriever settings, and a separate 500-query external-baseline evaluation, the central pattern is consistent: score-only selection and generic compression are useful but incomplete. Safe Adaptive Context and TACER preserve nearly all Fixed Top-10 quality while substantially reducing provider-reported token usage. Safe Adaptive is the more conservative con-

Table 8: Paired TACER–Adaptive- k (score-drop) confidence check in the main Llama-70B TF-IDF setting. Token deltas are TACER minus Adaptive- k (score-drop), so negative values mean TACER uses fewer tokens.

Dataset	$\Delta F1$ [95% CI]	F1 W/T/L	$\Delta Tokens$ [95% CI]	Token W/T/L
SciFact	+0.002 [-0.007,+0.011]	84/17/99	-204.4 [-254.8,-159.2]	155/1/44
BioASQ	-0.006 [-0.020,+0.009]	85/21/94	-261.2 [-328.0,-199.8]	149/0/51
HotpotQA	+0.137 [+0.087,+0.189]	40/152/8	+650.7 [+591.5,+707.1]	19/0/181
MS MARCO	-0.005 [-0.017,+0.007]	73/46/81	-76.0 [-93.5,-59.5]	134/1/65
ASQA	+0.036 [+0.021,+0.050]	122/16/62	+136.5 [+89.4,+191.7]	42/1/157

Table 9: Paired TACER–Safe Adaptive comparison in the main Llama-70B TF-IDF evaluation. Deltas are TACER minus Safe Adaptive; negative token deltas favor TACER. Intervals are percentile 95% CIs from 20,000 paired bootstrap resamples (seed 20260908).

Dataset	$\Delta F1$ [95% CI]	$\Delta Tokens$ [95% CI]
SciFact	+0.006 [-0.001,+0.013]	-373.2 [-521.1,-246.7]
BioASQ	-0.009 [-0.020,+0.003]	-213.9 [-307.1,-128.6]
HotpotQA	+0.000 [+0.000,+0.000]	-4.0 [-12.0,+0.1]
MS MARCO	-0.001 [-0.013,+0.012]	-216.9 [-270.5,-163.3]
ASQA	+0.002 [-0.003,+0.007]	-13.6 [-43.9,+16.0]

Table 10: Average behavior across five datasets. Retention is relative to Fixed Top-10 under the same retriever.

Retriever	Method	F1 Ret.	Token Red.
TF-IDF	Score-drop	93.1%	67.6%
TF-IDF	Safe Adaptive	98.4%	54.7%
TF-IDF	TACER	98.4%	61.5%
TF-IDF+CE	Score-drop	97.6%	60.2%
TF-IDF+CE	Safe Adaptive	99.7%	60.3%
TF-IDF+CE	TACER	98.8%	58.0%
Dense+CE	Score-drop	98.1%	58.2%
Dense+CE	Safe Adaptive	99.8%	58.1%
Dense+CE	TACER	99.8%	56.1%

troller; TACER trades some quality protection for greater token reduction in some settings. Neither policy universally dominates across retrievers and datasets.

Limitations

The evaluation is controlled rather than exhaustive: final runs use prepared splits, some dataset conversions are simplified, and TACER uses interpretable heuristic routing rather than learned calibration. The component ablation is limited to SciFact, so it does not isolate each routing component across task types. The targeted human annotation is a grounding check rather than a full human evaluation; a larger adjudicated study would better quantify factuality differences across methods. It is also deliberately stress-oriented, focusing on

Table 11: Two-annotator blinded annotation on 30 HotpotQA/ASQA queries. Counts use a conservative merge: correct only when both annotators mark correct; acceptable = correct or partially correct.

Method	Corr.	Part.	Inc./NEI	Accept.
Fixed Top-10	16	8	6	80.0%
Adaptive- k (paper)	9	9	12	60.0%
Safe Adaptive	19	9	2	93.3%
TACER	19	10	1	96.7%

HotpotQA and ASQA cases where context under-selection is most likely to matter, rather than estimating average accuracy over all RAG workloads.

Automatic metrics are imperfect, especially for biomedical and long-form answers, which is why we report them jointly with human labels instead of treating any single score as decisive. Retention percentages should also be read alongside absolute scores: the Fixed Top-10 F1 baselines are low on SciFact and MS MARCO, so small absolute changes can appear comparatively large when expressed as retention. For grounded generation, the relevant question is not only whether the answer string overlaps a reference, but whether the selected evidence is sufficient to support the answer under a chosen risk budget.

The method also assumes that the retrieved candidate list contains enough evidence to answer the query. TACER can decide whether to expose compact spans, selected documents, or broader fallback context, but it cannot recover evidence that retrieval never found. Its role is therefore a context-allocation layer to combine with stronger retrieval, factual verification, uncertainty estimation, and deployment-specific risk policies. Future work should evaluate learned or leave-one-dataset-out calibrated routers, larger adjudicated human studies, and production-style latency and call-count trade-offs.

Ethical Considerations

Because SciFact and BioASQ are biomedical, outputs should not be treated as medical advice or clinical evidence. Token-efficient RAG can reduce compute and cost, but deployments should monitor whether compression removes useful evidence.

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A Implementation Details

A.1 Dataset Sources and Evaluation Splits

The evaluation uses SciFact (Wadden et al., 2020), BioASQ (Tsatsaronis et al., 2015), HotpotQA (Yang et al., 2018), MS MARCO (Nguyen et al., 2016), and ASQA (Stelmakh et al., 2022). The final main evaluation uses 50 calibration examples and 200 held-out evaluation examples per dataset (1,000 evaluation queries total). The separate external-baseline evaluation uses the first 100 entries of each fixed 200-query evaluation file (500 total), selected deterministically from index 0 by the released batch script. Each file records stable query identifiers, and all methods within an evaluation use the same query set.

A.2 Model and Checkpoint Details

The hosted main-generator runs use meta-llama/Llama-3.3-70B-Instruct at temperature 0. The smaller-model robustness runs use the local Ollama model under identifier mistral, also at temperature 0. Its unchanged local manifest identifies a 7.2B-parameter GGUF model with Q4_K_M quantization, a 32,768-token context window, and model-layer digest sha256:f5074b1221da0f5a2910d33b642efa5b9eb58cfdddca1c79e16d7ad28aa2b31f. TF-IDF is the transparent first-stage retriever. Cross-encoder reranking uses cross-encoder/ms-marco-MiniLM-L-6-v2, and dense retrieval uses intfloat/multilingual-e5-large-instruct followed by the same cross-encoder. Reported semantic scores use sentence-transformers/all-MiniLM-L6-v2; the saved scores differ from the implementation’s lexical fallback values. Future runs should require the semantic model and record its revision explicitly.

A.3 SciFact Reference Construction

SciFact supplies claims, gold documents, and relevance labels rather than short natural-language reference answers. We therefore generated an auxiliary reference for each query with openai/gpt-oss-120b through the Berget AI OpenAI-compatible API at temperature 0. The input contains only the SciFact claim and its gold evidence. The system message is “You create concise reference answers for factual RAG evaluation.” The user template is:

Create one short reference answer for evaluating a RAG system on SciFact. The input is a scientific claim and the gold evidence document text. Decide whether the evidence supports, contradicts, or is insufficient for the claim.

Rules: Write one concise natural-language reference answer. Match the existing SciFact reference style: usually 4 to 20 words, and never more than 30 words. If the evidence supports the claim, state the supported claim directly. If the evidence contradicts the claim, begin with “The evidence contradicts the claim” and briefly say why. If the evidence is insufficient, write exactly: “The evidence is insufficient.” Do not add citations, bullet points, or extra explanation.

Claim: *[claim]*

Gold evidence: *[gold evidence]*

These references are used only for answer-level F1, coverage, and semantic similarity; retrieval metrics continue to use the original SciFact relevance labels. The available artifacts do not record an independent expert validation of every generated reference, so answer-level SciFact results are auxiliary evidence alongside the four datasets with source-derived references.

A.4 Answer-Generation Prompts

All context policies for a dataset use the same answer-generation prompt; methods differ only in the evidence inserted into it. SciFact, BioASQ, MS MARCO, and ASQA use the following default template:

Answer the question using only the provided documents. If the documents do not contain enough evidence, say that the evidence is insufficient.

Question: *[query]*

Documents: *[numbered document blocks]*

Answer:

Each block is formatted as [Document *i* | document-id] followed by its full or compressed text. HotpotQA uses the concise template because its references are short exact answers:

Use only the evidence below to answer the question. Write only the final answer. Use as few words as possible, usually 1 to 5 words. Use the same key terms as the evidence when possible. Do not explain your reasoning. Do not repeat the question. If the evidence does not answer the question, write exactly: insufficient evidence.

Question: *[query]*

Evidence: *[numbered document blocks]*

Short answer:

A.5 Metric and Budget Definitions

Answer coverage is the fraction of unique reference-answer tokens recovered by the generated answer:

$$\text{coverage}(a, r) = \frac{|T(a) \cap T(r)|}{|T(r)|},$$

where $T(\cdot)$ is the normalized token set used by the evaluation script. It is therefore a recall-style lexical metric, complementary to token F1 and semantic similarity.

The configured maximum output-token budget is 80 for SciFact, BioASQ, HotpotQA, and MS MARCO, and 160 for ASQA. The TACER routing rule that checks whether an answer is a long-answer setting therefore uses the > 120 output-token threshold only for ASQA in the reported configuration.

A.6 Calibration and Compression Settings

The 50-example calibration split is used only to choose lightweight routing thresholds and budget behavior. Thresholds are selected by inspecting calibration-set score gaps, normalized entropy, query length, lexical evidence coverage, and fallback behavior, then choosing conservative cut-offs that preserve Fixed Top-10 quality on distributed or long-answer cases while allowing aggressive compression on concentrated-evidence cases. Dataset identifiers are not used as policy inputs.

The reported TACER and Safe Adaptive compression is the lexical sentence-neighborhood compression described below, not an LLMLingua-2 call. LLMLingua-2 appears only in the external-baseline evaluation, where the implementation uses the microsoft/llmlingua-2-*xl*-roberta-large-meetingbank checkpoint with its budget chosen to match each routed context budget as closely as possible before answer generation. We report this tested configuration rather than a full LLMLingua-2 quality–efficiency sweep.

A.7 Retriever Details

For a corpus of N documents, inverse document frequency is computed as:

$$\text{idf}(t) = \log \left(\frac{1 + N}{1 + \text{df}(t)} \right) + 1,$$

where $\text{df}(t)$ is the number of documents containing term t . Query and document vectors are com-

Table 12: TACER routing rules used in the final implementation.

Condition	Routing action
Concise prompt with bridge evidence, multi-hop shape, or distributed complexity	Use safer Safe Adaptive ladder
Long-answer setting with > 120 output tokens and non-focused complexity	Use safer Safe Adaptive ladder
Claim-style prompt with at most 120 output tokens	Use compact coverage-guided evidence
Distributed complexity with a non-concentrated ranking	Use safer Safe Adaptive ladder
Focused or concentrated evidence otherwise	Use compact coverage-guided evidence

puted in the same TF-IDF space, and documents are ranked by cosine similarity:

$$\text{sim}(q, d) = \frac{q \cdot d}{\|q\| \|d\|}.$$

A.8 Evidence Compression Scoring

For compact evidence modes, candidate sentences are scored as:

$$\begin{aligned} \text{score}(q, s, i) = & .40 O(q, s) + .25 R(q, s) \\ & + .20 P(q, s) + .10 D(q, s) \\ & + .05(1 + i)^{-1}. \end{aligned}$$

Here O , R , P , and D denote unigram overlap, rare query-term coverage, phrase overlap, and token density. The position bonus mildly favors earlier sentences.

B Routing and Fallback Details

TACER uses query text, retrieval scores, lexical evidence coverage, and the configured answer-length budget; it does not use dataset identifiers or an additional LLM call before generation. The query-shape detector labels concise prompts as factoid unless the query contains multi-hop cues such as *which*, *where*, *when*, *whose*, *both*, or *and* and has at least six content words. Non-concise questions are treated as factoid-style questions when they contain a question mark, and otherwise as claim-style inputs.

Evidence complexity is assigned from general query features. A concise query is treated as distributed when it contains connective evidence or at least two capitalized entities and has at least six content words. Otherwise, any query with at least twelve content words, two connective cues, three capitalized entities, or two numeric expressions is treated as distributed; a query with at least seven content words or one capitalized entity is moderate; remaining queries are focused.

For concise short-answer prompts, fallback is triggered only for an empty answer or for an answer with more than 18 content words and answer-context overlap below 0.80. These concise-task rules are intentionally conservative because many

Table 13: Fallback risk rules for evidence-style prompts. Fallback is triggered when total risk is at least 3.

Condition	Risk increment
Empty answer	10
Contains weak-answer phrase	+3
Answer-context overlap < 0.08	+3
Context-query coverage < 0.35	+2
Fewer than 5 answer content words and answer-context overlap < 0.20	+2
Answer-query overlap < 0.04	+1
More than 24 answer content words and answer-context overlap < 0.90	+1

correct answers are names, dates, places, or short phrases with limited lexical overlap.

C Additional Per-Dataset Comparisons

Table 14 adds the requested fixed-depth operating points to the main TF-IDF comparison. Each entry reports answer F1 and token reduction relative to Fixed Top-10 for the same dataset.

C.1 External Baselines by Dataset

Table 15 provides the dataset-level values behind the macro-averages in Table 6.

D Generator Robustness

Table 16 reports the smaller-model check without repeating the complete method ladder.

E Retriever-Robustness Details

Table 17 reports dataset-level differences between TACER and Adaptive- k (score-drop) across retrieval settings.

F Representative Failure Examples

G Human Annotation Protocol

The metric-validation study contains 120 outputs from SciFact, HotpotQA, and BioASQ sampled across Llama-70B and Mistral. Two annotators agreed on 84 examples, yielding 70.0% raw agreement and Cohen’s $\kappa = 0.477$; among agreed labels, 65 are correct, 6 partially correct, and 13 incorrect. The dataset-level agreed counts are: SciFact, 20 correct, 2 partially correct, and 5 incorrect; HotpotQA, 26 correct, 0 partially correct, and 4 incorrect; and BioASQ, 19 correct, 4 partially correct, and 4 incorrect.

The targeted controller study contains 15 HotpotQA and 15 ASQA queries. For each query, annotators see four saved Llama-70B outputs in randomized A/B/C/D order: Fixed Top-10, Adaptive- k (paper), Safe Adaptive Context, and TACER.

They see the query, reference, answer, and retrieved evidence, while method identities, automatic scores, and token counts remain hidden. Two annotators label all 120 outputs as correct, partially correct, incorrect, or not enough information. Exact four-way agreement is 68.3% ($\kappa = 0.422$); binary acceptable/unacceptable agreement is 92.5% ($\kappa = 0.688$). Table 19 reports the conservative merge, where an answer is correct only if both annotators mark it correct.

Table 14: Per-dataset Llama-70B TF-IDF operating points. Each cell reports F1 / token reduction.

Dataset	Fixed Top-3	Fixed Top-5	Fixed Top-10	Score-drop	Safe Adaptive	TACER
SciFact	0.245 / 66.7%	0.245 / 47.7%	0.243 / 0.0%	0.251 / 74.9%	0.247 / 69.9%	0.253 / 80.9%
BioASQ	0.322 / 66.7%	0.326 / 47.5%	0.328 / 0.0%	0.319 / 73.2%	0.323 / 74.6%	0.314 / 80.8%
HotpotQA	0.532 / 67.7%	0.555 / 49.9%	0.645 / 0.0%	0.490 / 70.4%	0.627 / 32.3%	0.627 / 32.6%
MS MARCO	0.197 / 62.2%	0.198 / 43.8%	0.196 / 0.0%	0.199 / 48.6%	0.195 / 38.5%	0.195 / 54.1%
ASQA	0.398 / 64.6%	0.416 / 45.9%	0.422 / 0.0%	0.368 / 70.9%	0.402 / 58.3%	0.404 / 59.4%

Table 15: External-baseline results covering 100 queries from each dataset (500 total). Token reduction is relative to Fixed Top-10 within each dataset.

Dataset	Method	F1	Semantic	Tokens	Token Red.
SciFact	Fixed Top-10	0.231	0.497	3388	0.0%
SciFact	Adaptive- k (paper)	0.238	0.496	2451	27.7%
SciFact	Top-10 + LLMLingua-2	0.246	0.508	929	72.6%
SciFact	Adaptive- k + LLMLingua-2	0.240	0.501	711	79.0%
SciFact	FLARE-lite	0.247	0.501	1175	65.3%
SciFact	Safe Adaptive Context	0.236	0.490	899	73.5%
SciFact	TACER	0.240	0.501	703	79.3%
BioASQ	Fixed Top-10	0.333	0.751	3425	0.0%
BioASQ	Adaptive- k (paper)	0.336	0.760	2469	27.9%
BioASQ	Top-10 + LLMLingua-2	0.261	0.709	847	75.3%
BioASQ	Adaptive- k + LLMLingua-2	0.263	0.710	654	80.9%
BioASQ	FLARE-lite	0.315	0.720	580	83.1%
BioASQ	Safe Adaptive Context	0.329	0.752	788	77.0%
BioASQ	TACER	0.319	0.736	651	81.0%
HotpotQA	Fixed Top-10	0.647	0.750	1685	0.0%
HotpotQA	Adaptive- k (paper)	0.591	0.707	1199	28.9%
HotpotQA	Top-10 + LLMLingua-2	0.424	0.566	494	70.7%
HotpotQA	Adaptive- k + LLMLingua-2	0.401	0.564	379	77.5%
HotpotQA	FLARE-lite	0.576	0.698	1176	30.2%
HotpotQA	Safe Adaptive Context	0.633	0.728	1163	31.0%
HotpotQA	TACER	0.633	0.729	1155	31.4%
ASQA	Fixed Top-10	0.436	0.791	1195	0.0%
ASQA	Adaptive- k (paper)	0.433	0.783	875	26.8%
ASQA	Top-10 + LLMLingua-2	0.318	0.693	357	70.1%
ASQA	Adaptive- k + LLMLingua-2	0.321	0.692	285	76.1%
ASQA	FLARE-lite	0.342	0.686	275	77.0%
ASQA	Safe Adaptive Context	0.414	0.764	498	58.3%
ASQA	TACER	0.415	0.763	516	56.8%
MS MARCO	Fixed Top-10	0.181	0.466	1408	0.0%
MS MARCO	Adaptive- k (paper)	0.183	0.465	1221	13.3%
MS MARCO	Top-10 + LLMLingua-2	0.152	0.446	484	65.6%
MS MARCO	Adaptive- k + LLMLingua-2	0.153	0.445	429	69.6%
MS MARCO	FLARE-lite	0.197	0.440	332	76.4%
MS MARCO	Safe Adaptive Context	0.191	0.463	910	35.4%
MS MARCO	TACER	0.189	0.449	655	53.5%

Table 16: Generator comparison in the TF-IDF setting. Each cell reports F1 / token reduction relative to Fixed Top-10 for that generator and dataset.

Dataset	Method	Llama-70B	Mistral 7B
SciFact	Score-drop	0.251 / 74.9%	0.259 / 74.0%
	Safe Adaptive	0.247 / 69.9%	0.242 / 71.7%
	TACER	0.253 / 80.9%	0.246 / 79.4%
BioASQ	Score-drop	0.319 / 73.2%	0.318 / 72.1%
	Safe Adaptive	0.323 / 74.6%	0.305 / 73.2%
	TACER	0.314 / 80.8%	0.316 / 79.8%
HotpotQA	Score-drop	0.490 / 70.4%	0.282 / 71.4%
	Safe Adaptive	0.627 / 32.3%	0.369 / 30.9%
	TACER	0.627 / 32.6%	0.374 / 31.1%
MS MARCO	Score-drop	0.199 / 48.6%	0.212 / 50.1%
	Safe Adaptive	0.195 / 38.5%	0.217 / 35.1%
	TACER	0.195 / 54.1%	0.209 / 54.1%
ASQA	Score-drop	0.368 / 70.9%	0.351 / 72.6%
	Safe Adaptive	0.402 / 58.3%	0.359 / 61.9%
	TACER	0.404 / 59.4%	0.359 / 61.9%

Table 17: Direct comparison between TACER and Adaptive- k (score-drop). Positive F1 deltas favor TACER; positive reduction deltas mean TACER saves more tokens.

Retriever	Dataset	Adaptive- k F1	TACER F1	TACER F1 Delta	TACER Reduction Delta
TF-IDF	SciFact	0.251	0.253	+0.002	+6.0%
TF-IDF	BioASQ	0.319	0.314	-0.006	+7.6%
TF-IDF	HotpotQA	0.490	0.627	+0.137	-37.8%
TF-IDF	MS MARCO	0.199	0.195	-0.005	+5.5%
TF-IDF	ASQA	0.368	0.404	+0.036	-11.5%
TF-IDF + CE	SciFact	0.262	0.251	-0.011	+5.6%
TF-IDF + CE	BioASQ	0.336	0.320	-0.015	+11.5%
TF-IDF + CE	HotpotQA	0.560	0.676	+0.116	-34.1%
TF-IDF + CE	MS MARCO	0.204	0.200	-0.005	+7.4%
TF-IDF + CE	ASQA	0.415	0.418	+0.003	-1.3%
Dense + CE	SciFact	0.264	0.255	-0.009	+5.5%
Dense + CE	BioASQ	0.338	0.324	-0.013	+11.5%
Dense + CE	HotpotQA	0.573	0.705	+0.131	-34.9%
Dense + CE	MS MARCO	0.204	0.200	-0.004	+7.7%
Dense + CE	ASQA	0.415	0.417	+0.002	-0.4%

Table 18: Representative examples from the main Llama-70B TF-IDF run. Metrics are F1 / coverage / semantic similarity.

Pattern	Query	Score-drop	TACER
Score-only under-selection	HotpotQA: birth date of a Trump campaign consultant	0.000 / 0.000 / 0.017; 1 doc	1.000 / 1.000 / 1.000; 10 docs
Long-form evidence need	ASQA: who played the baby in <i>Baby's Day Out</i> ?	0.292 / 0.261 / 0.490; 1 doc	0.706 / 0.739 / 0.889; 3 docs
Compression success	SciFact: hypocretin neurons induce a panic-prone state	0.216 / 0.889 / 0.800; 3 docs	0.485 / 0.889 / 0.946; 4 docs
Over-compression risk	BioASQ: is pregabalin effective for sciatica?	0.661 / 0.794 / 0.944; 1 doc	0.250 / 0.176 / 0.813; 2 docs

Table 19: Targeted annotation results by dataset. Acceptable combines correct and partially correct.

Dataset	Method	Corr.	Part.	Inc./NEI	Accept.
HotpotQA	Fixed Top-10	10	1	4	73.3%
	Adaptive- k (paper)	4	0	11	26.7%
	Safe Adaptive	12	2	1	93.3%
	TACER	12	2	1	93.3%
ASQA	Fixed Top-10	6	7	2	86.7%
	Adaptive- k (paper)	5	9	1	93.3%
	Safe Adaptive	7	7	1	93.3%
	TACER	7	8	0	100.0%

Table 20: Alignment between automatic metric buckets and agreed human labels in the metric-validation study.

Metric	Bucket	Items	Correct	Partially Correct	Incorrect
Token F1	Low (0.00–0.33)	33	48.5%	18.2%	33.3%
Token F1	Medium (0.34–0.66)	22	90.9%	0.0%	9.1%
Token F1	High (0.67–1.00)	29	100.0%	0.0%	0.0%
Semantic	Low (0.00–0.49)	23	47.8%	13.0%	39.1%
Semantic	Medium (0.50–0.74)	12	58.3%	25.0%	16.7%
Semantic	High (0.75–1.00)	49	95.9%	0.0%	4.1%
Coverage	Low (0.00–0.33)	17	17.6%	29.4%	52.9%
Coverage	Medium (0.34–0.66)	14	71.4%	7.1%	21.4%
Coverage	High (0.67–1.00)	53	98.1%	0.0%	1.9%